

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			large surface area/ large area for absorption	1			1		
	(b)			Higher concentration of water outside the cell /lower concentration of solutes outside (1) Water moves into the cell (down a concentration gradient)(1) Through a selectively permeable membrane (1) via osmosis (1)	4			4		
	(c)			Indicative content: • lack of oxygen therefore less aerobic respiration • less energy available • less ATP produced; • active transport of mineral salts • transporting from low to high concentration • Requires ATP/ energy • there is mineral/ nutrient deficiency • poor growth - lack of nitrates • yellow leaves - lack of potassium 5-6 marks All mineral deficiencies should be accounted for as well as linking oxygen lack of oxygen with less ATP from active transport needed for 6 marks. Use of energy rather than ATP gains 5 marks <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	2	4		6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				3-4 marks Description of one of two mineral deficiencies, linking aerobic respiration with lack of oxygen. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks • there is mineral deficiency • poor growth - lack of nitrates OR yellow leaves - lack of potassium Reference to mineral deficiency and one plant nutrient gains 2 marks. No mention of specific nutrient gains 1 mark. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks: No attempt made or no response worthy of credit.						
				Question 8 total	7	4	0	11	0	0